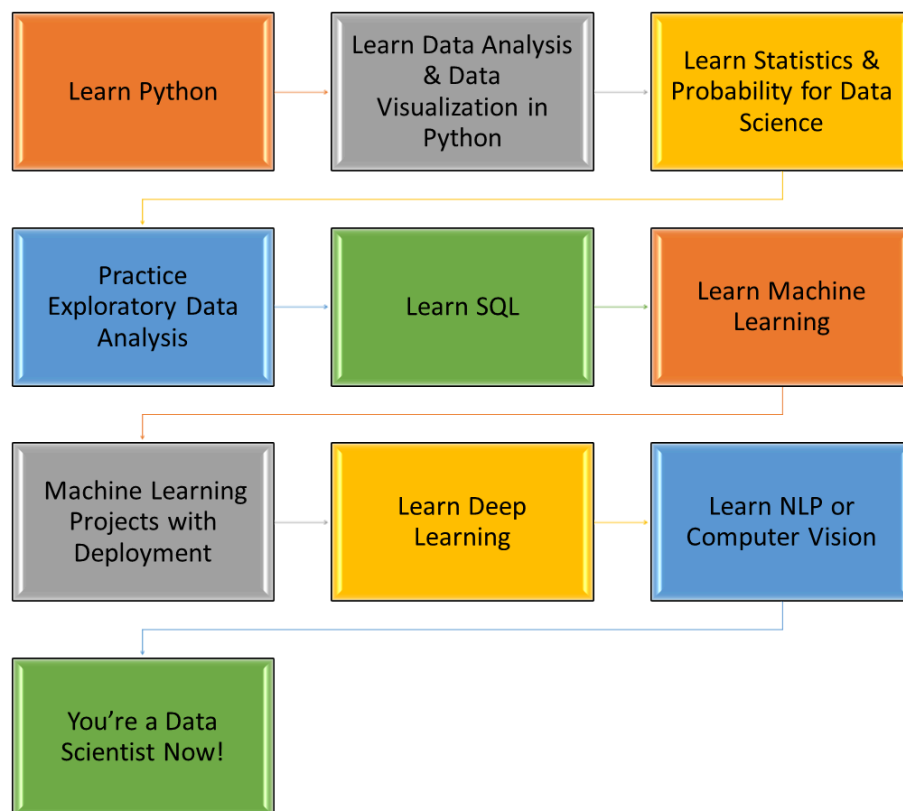


Detailed Roadmap & Free Resources to Learn Data Science

This roadmap is designed for complete beginners to acquire Data Science skills, even without any prior coding or computer science knowledge. It covers all technical skills with free learning resources provided for each.

Expected Duration to learn Data Science: 6 to 9 months

How to Become a Data Scientist



Step 1 - Learn Python

Topics you need to cover in Python for Data Science

- Variables, Numbers, Strings
- Lists, Dictionaries, Sets, Tuples
- If condition, for loop
- Functions, Lambda Functions
- Modules (pip install)
- Read, Write files

- Exception handling
- Classes, Objects

YouTube Videos to learn Python

- https://youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU&si=9ING_XKiJKrSYDgR
- https://youtube.com/playlist?list=PLu0W_9lI9agwh1XjRt242xIpHhPT2llg&si=uY9q2ceGXl6bmQl4

Free Courses to learn Python

- <https://www.edx.org/learn/python/harvard-university-cs50-s-introduction-to-programming-with-python>
- <https://www.geeksforgeeks.org/courses/python-course-certification-free>
- <https://www.freecodecamp.org/learn/scientific-computing-with-python/>

Exercises to learn Python

- <https://github.com/codebasics/py/tree/master/Basics/Exercise>

Step 2 - Learn Data Analysis & Data Visualization in Python

For Data Analysis in Python, learn Numpy & Pandas

- a. YouTube videos to learn Numpy (You can choose anyone from the below playlists)
 - <https://www.youtube.com/playlist?list=PLeo1K3hjS3uset9zIVzJWqplaWBiacTEU>
 - <https://www.youtube.com/watch?v=ZaKzw9tULeM&list=PLjVLYmrlmjGfgBKkIFBkMNGG7qyRfo00W>
 - <https://youtube.com/playlist?list=PL9n0l8rSshSnragNblKDBsT8Xu3otp3jA&si=ltkXQ5eAegFYbkWk>
- b. YouTube videos to learn Pandas (You can choose anyone from the below playlists)
 - <https://youtube.com/playlist?list=PLeo1K3hjS3uuASpe-1LjfG5f14Bnozjwy&si=0HgEHxoGYWpbSPcS>
 - <https://youtube.com/playlist?list=PLjVLYmrlmjGdEE2jFpL71LsVH5QjDP5s4&si=K5UEPCkXoIFWusN8>

For Data Visualization in Python, learn Matplotlib & Seaborn

- a. YouTube videos to learn Matplotlib (You can choose anyone from the below playlists)
 - https://youtube.com/playlist?list=PLeo1K3hjS3uu4Lr8_kro2AqaO6CFYgKOI&si=ejpRnZcF1JQhJUtk
 - https://youtube.com/playlist?list=PLjVLYmrlmjGcC0B_FP3bkJ-JIPkV5GuZR&si=klf5Fi7QaNyRZVqT
- b. YouTube videos to learn Seaborn

- <https://youtube.com/playlist?list=PLjVLYmrlmjGfhqSO3rF4n02rrj9w2Ch2C&si=sUuG7RZj6c1VaX47>

You can also refer to the below one-shot videos, which cover all the concepts related to Python for Data Science

- <https://www.youtube.com/watch?v=LHBE6Q9Xlzl&t=2s>
- <https://www.youtube.com/watch?v=JDcZBzb46ts>
- <https://www.youtube.com/watch?v=GPVsHOIRBBI> (Only for Data Analysis & Visualization in Python)

Step 3 - Learn Probability & Statistics for Data Science

Topics you need to cover in Statistics & Probability for Data Science

- Basics: Descriptive vs inferential statistics, continuous vs discrete data, nominal vs ordinal data
- Basic plots: Histograms, pie charts, bar charts, scatter plots, etc. ▪ Measures of central tendency: mean, median, mode
- Measures of dispersion: variance, standard deviation
- Probability basics
- Distributions: Normal distribution
- Correlation and covariance
- Central limit theorem
- Hypothesis testing: p-value, confidence interval, type 1 vs type 2 error, Z test, t-test, ANOVA

YouTube Videos to learn Statistics & Probability for Data Science (You can choose anyone from the below playlists)

- <https://www.youtube.com/watch?v=LZzq1zSL1bs>
- <https://www.youtube.com/watch?v=Vfo5le26lhY>
- <https://www.youtube.com/watch?v=bLZ-LSsQMcc>

Now, after learning Python, Data Analysis, Data Visualization using Python, & Statistics for Data Science, you can use all these to do Exploratory Data Analysis, which is the first step in building a model.

Step 4 - Practice Exploratory Data Analysis

Watch anyone of the videos below to understand how to do EDA -

- https://www.youtube.com/watch?v=fFi_TBw27is&list=PL9n0l8rSshSnZHFChXIG5Q9f0UN8O-UYX&index=3
- [youtube.com/watch?v=fHFOANOHwh8](https://www.youtube.com/watch?v=fHFOANOHwh8)

Use Kaggle to find Datasets for Exploratory Data Analysis (Practice EDA on at least 5 datasets) - <https://www.kaggle.com/datasets?search=Exploratory+Data+Analysis>

Step 5 - Learn SQL

SQL is extremely important for data science. It allows you to work with structured data stored in databases. As a data scientist, you need SQL to extract, manipulate, and analyze data from these databases.

YouTube videos to learn SQL

- <https://www.youtube.com/watch?v=nopIGY1zJE0>
- <https://www.youtube.com/watch?v=2w-M4z-iQpk>

Free Courses to learn SQL

- <https://www.w3schools.com/sql/>
- <https://sqlbolt.com/>

Step 6 - Learn Machine Learning (Most important)

Topics to cover:

Machine Learning: Preprocessing

- Handling NA values, outlier treatment, data normalization
- One hot encoding, label encoding
- Feature engineering
- Train test split
- Cross-validation

Machine Learning: Model Building

- Types of ML: Supervised, Unsupervised
- Supervised: Regression vs Classification
- Linear models
 - Linear regression, logistic regression
 - Gradient descent
- Nonlinear models (tree-based models)
 - Decision tree
 - Random forest
 - XGBoost
- Model evaluation

- Regression: Mean Squared Error, Mean Absolute Error, MAPE
Classification: Accuracy, Precision-Recall, F1 Score, ROC Curve, Confusion matrix
Hyperparameter tuning: GridSearchCV, RandomSearchCV
→ Unsupervised: K means, Hierarchical clustering, Dimensionality reduction (PCA)

Youtube Videos to learn Machine Learning

- <https://youtube.com/playlist?list=PLeo1K3hjS3uvCeTYTeyfe0-rN5r8zn9rw&si=WZA6ITnDR2-YPEfj>
- <https://www.youtube.com/watch?v=NWONeJKN6kc>
- <https://www.edx.org/learn/machine-learning/harvard-university-data-science-machine-learning>

Step 7 - Machine Learning Projects with Deployment

Do end-to-end ML projects covering both Regression & Classification

For Regression

<https://youtube.com/playlist?list=PLeo1K3hjS3uu7clOTwsp94PcHbzqpAdg&si=tQNxP-xopMIEa0CI>

For Classification

<https://youtube.com/playlist?list=PLeo1K3hjS3uvaRHZLI-jLovljBP14QTXc&si=5E2O2XT5VaccMWK-a>

ML Projects

- https://youtube.com/playlist?list=PLKnIA16_RmvY5eP91BGPao0vXUYmldtfPQ&si=FLiysiVRxYp_xkgt
- https://youtube.com/playlist?list=PLfFghEzKVmjvuSA67LszN1dZ-Dd_pkus6&si=hb4jc_0sr4W80Gln

You can also find ML projects on the Kaggle Platform - <https://www.kaggle.com/>

Step 8 - Learn Deep Learning

Topics to Cover

- What is a neural network?
- Building multilayer perceptron
- Special neural network architectures
 - Convolutional neural network (CNN)
 - Sequence models: RNN, LSTM

YouTube Playlists

- https://youtube.com/playlist?list=PLeo1K3hjS3uu7CxAcxVndI4bE_o3BDtO&si=OJA g9Mfb-hfUapM
- https://youtube.com/playlist?list=PLOrU905yPYXJsJSHJsiE779KfcrRCgz4v&si=_yp3 K8IOEpAAqcS5
- https://youtube.com/playlist?list=PLKnIA16_RmvYuZauWaPIRTC54KxSNLtNn&si=zd IKOR8dhs26vJxj

Projects

- <https://youtube.com/playlist?list=PLeo1K3hjS3utJFNGyBplvjWgSDY0eOE8S&si=aP Z4if0tzQYI8fok>
- <https://youtube.com/playlist?list=PLfFghEzKVmjvyT7wdfDJSHhS0FdF8jd-U&si=DfB U6k28idQuG5iU>

Step 9 - Learn NLP or Computer Vision

Choose a specialized track, which is either NLP or Computer vision. No need to learn both.

NLP Youtube Playlist -

<https://youtube.com/playlist?list=PLeo1K3hjS3uuvuAXhYjV2IMEShq2UYSwX&si=59s20mkb B5-7RYwP>

CV Youtube playlist -

<https://youtube.com/playlist?list=PLzWRmD0Vi2KVsrCqA4VnztE4t71KnTnP5&si=f0Rcf0dib RCsUMmS>

Congratulations, You're a Data Scientist now!

Data Science Projects & Certifications you can add to your Resume

Data Science Projects to add to your Resume

- <https://github.com/krishnaik06/Data-Science-Projects-For-Resumes>
- <https://github.com/vab-101/Data-Science-Projects>
- <https://github.com/durgeshsamariya/Data-Science-Machine-Learning-Project-with-Source-Code>
- <https://github.com/SUKHMAN-SINGH-1612/Data-Science-Projects>

Top Data Science Certifications you can add to your resume (not compulsory) -

<https://www.geeksforgeeks.org/top-7-data-science-certifications-that-you-can-consider/>